

3. Write a program to implement all activation functions in neural network.

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import numpy as np

# 1. Binary Step Function
def binary_step(x):
    return np.where(x >= 0, 1, 0)

# 2. Sigmoid Function
def sigmoid(x):
    return 1 / (1 + np.exp(-x))

# 3. Tanh Function
def tanh(x):
    return np.tanh(x)

# 4. ReLU Function
def relu(x):
    return np.maximum(0, x)

# 5. Leaky ReLU Function
def leaky_relu(x, alpha=0.01):
    return np.where(x > 0, x, alpha * x)

# 6. Softmax Function
def softmax(x):
    e_x = np.exp(x - np.max(x)) # for numerical stability
    return e_x / e_x.sum(axis=0)

# -----
# Testing the functions
# -----

# Input array
x = np.array([-2, -1, 0, 1, 2])

print("Input:", x)

print("\nBinary Step:", binary_step(x))
print("Sigmoid:", sigmoid(x))
print("Tanh:", tanh(x))
print("ReLU:", relu(x))
print("Leaky ReLU:", leaky_relu(x))

# Softmax requires a different kind of input (usually probabilities)
x_soft = np.array([2.0, 1.0, 0.1])
print("\nSoftmax:", softmax(x_soft))

```